Ayush Maheshwari

Senior Solutions Architect, NVIDIA | Ph.D. [CSE, IIT Bombay] | Foundation Models

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Summary

Ph.D. in Computer Science and Engineering from IIT Bombay and Senior Solutions Architect at NVIDIA, working on foundation models, multilingual NLP, Indic LLMs, distributed training, and model evaluation. Previously led Indic LLM development from scratch at Vizzhy, including training on large accelerator cluster.

Skills

LLM Systems: Foundation Models, Distributed Training, LLM Pretraining, Post-training, Instruction Tuning, Model Adaptation, Inference Optimization and Model Deployment

NLP: Multilingual NLP, Low-Resource Languages, Neural Machine Translation, Weak Supervision, Semi-Supervised Learning

Programming Languages/Tools: Python, PyTorch, Megatron, Bash, SQL, NumPy, scikit-learn, C, C++, JavaScript

Experience

NVIDIA — **Gurugram, India**
Senior Solutions Architect, NVAITC — *Oct 2024 – Present*

- ◇ Working on foundation-model research across multilingual, domain-specific, and scientific AI problems, including data design, model adaptation, evaluation, and deployment on computing platforms.
- ◇ Collaborating with academic researchers, research organizations, and external AI teams on large-scale model development and scientific AI workloads.
- ◇ Reviewing large-scale training and inference setups, and debugging performance issues across data, model, compute, and serving layers.

Vizzhy Inc — **Bengaluru, India**
Lead Research Scientist, Language Model Team — *Sep 2023 – Sep 2024*

- ◇ Led a 5-member research team that built Indic large language models from scratch, covering data curation, tokenizer training, architecture choices, pretraining, post-training, and evaluation.
- ◇ Trained Indic LLMs on a 1,024 accelerator cluster, handling distributed training, experiment runs, checkpointing, failure recovery, throughput debugging, and model-quality analysis.

Additional Experience

Adobe Research — **Bengaluru, India**
Research Intern, Document Cloud Team — *May 2021 – Aug 2021*

NCETIS, IIT Bombay — **Mumbai, India**
Project Engineer — *May 2016 – Dec 2018*

Water & Climate Lab, IIT Gandhinagar — **Gandhinagar, India**
Junior Research Fellow — *May 2015 – Apr 2016*

Tata Consultancy Services — **Mumbai, India**
Systems Engineer — *Oct 2011 – Jun 2013*

Education

Indian Institute of Technology Bombay — **Mumbai, India**
Ph.D., Computer Science and Engineering — *2019 – 2024*
Thesis: Knowledge Integration in NLP Models using Constraint Ingestion and Generation. Advisor: Prof. Ganesh Ramakrishnan

Tata Institute of Social Sciences — **Mumbai, India**
M.Sc. — *2013 – 2015*

University College of Engineering, Rajasthan Technical University — **Kota, India**
B.Tech., Computer Science and Engineering — *2007 – 2011*

Selected Publications

- **AQBENCH: Benchmarking Neural Surrogates for Air Quality Forecasting**. NeurIPS, 2026.
- **IndicParam: Benchmark to evaluate LLMs on low-resource Indic languages**. EMNLP (Main), 2026.
- **LexGen: Domain-aware Multilingual Lexicon Generation**. ACL (Main), 2025.
- **DictDis: Dictionary Constrained Disambiguation for Improved NMT**. Findings of EMNLP, 2024.
- **Adaptive Mixing of Auxiliary Losses in Supervised Learning**. AAAI, 2023. Oral Presentation.
- **Learning to Robustly Aggregate Labeling Functions for Semi-supervised Data Programming**. Findings of ACL, 2022.
- **Data Programming using Semi-Supervision and Subset Selection**. Findings of ACL, 2021.
- **Joint Learning of Hyperbolic Label Embeddings for Hierarchical Multi-label Classification**. EACL, 2021.

Open Source, Awards & Service

- **SPEAR**: Python library for data programming and weak supervision to create training data. [GitHub]
- **UDAAN**: ML-assisted post-editing workbench for producing publishable translations in Indian languages. [Website]
- **Impactful Research Award 2023**, IIT Bombay, for Ph.D. thesis work.
- **Best Demo Paper Award**, CODS-COMAD 2023, for UDAAN.
- **Reviewer**: ARR, ACL, EMNLP, and EACL.
- **Co-Founder**, Temples of India: web and app platform listing 700K+ temples across the Indian subcontinent. [Website]

Selected Talks

- **Building Data-Efficient Large Language Models**. Masterclass, NASSCOM NTLF, 2024.
- **Data-Efficient Machine Learning for End-to-End Educational Content Creation**. Invited Tutorial, International Conference on Educational Data Mining, 2023.
- **SPEAR: Data Programming using Semi-Supervision**. DECILE Workshop, 2022. [Talk]